

GLENCOE, MN, USA

WMO: 727517

Lat: 44.756N

Lon: 94.081W

Elev: 302

StdP: 97.74

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-22.0	-28.5	0.3	-23.5	-26.1	0.4	-21.4	11.9	-7.5	11.0	-8.3	2.5	300	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.1	23.3	30.2	22.1	28.8	21.3	25.1	29.9	23.9	28.3	22.9	27.0	4.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.1	28.1	22.5	17.8	26.9	21.4	16.6	25.8	78.2	29.7	73.2	28.3	69.2	27.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.8	7.8	DB	-27.8	35.1	3.6	1.4	-30.4	36.1	-32.5	36.9	-34.6	37.6	-37.2	38.6	(4)
(5)			WB	-28.1	26.9	3.5	1.3	-30.6	27.9	-32.7	28.7	-34.6	29.4	-37.2	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-9.1	-7.4	0.2	8.1	14.5	20.2	22.5	20.7	17.0	9.0	1.8	-6.2	(6)	
	DBStd	12.36	7.28	6.88	7.11	5.61	5.00	3.45	2.99	3.08	4.68	5.23	5.97	6.60	(7)	
	HDD10.0	2344	592	487	311	100	14	0	0	0	4	83	253	501	(8)	
	HDD18.3	4247	850	720	562	309	141	19	4	12	81	294	496	759	(9)	
	CDD10.0	1501	0	0	8	43	153	307	387	332	213	52	7	0	(10)	
	CDD18.3	362	0	0	1	2	22	76	132	86	40	4	0	0	(11)	
	CDH23.3	3176	0	0	2	32	247	653	1174	663	352	52	1	0	(12)	
	CDH26.7	939	0	0	0	8	82	196	373	169	98	12	0	0	(13)	
(14) Wind	WSAvg	3.2	3.6	3.6	3.6	4.0	3.6	2.9	2.4	2.2	2.8	3.4	3.6	3.4	(14)	
(15) Precipitation	PrecAvg	762	16	16	39	68	94	121	96	105	78	58	46	19	(15)	
	PrecMax	995	37	29	98	179	176	230	206	193	177	151	129	53	(16)	
	PrecMin	528	1	1	6	12	28	63	33	11	11	10	3	6	(17)	
	PrecStd	125	10	9	23	36	39	44	46	46	45	39	38	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	11.6	20.9	27.2	32.6	33.0	33.8	32.5	32.2	28.0	20.4	9.9	(19)	
		MCWB	4.0	10.0	14.9	14.8	19.6	22.8	25.5	24.1	22.4	17.7	13.7	8.0	(20)	
	2%	DB	3.3	6.2	16.4	22.7	28.9	31.0	32.1	30.1	29.2	24.1	16.2	6.0	(21)	
		MCWB	1.8	3.6	11.8	12.2	17.9	21.7	24.7	23.4	21.0	15.4	11.4	3.8	(22)	
	5%	DB	2.1	3.5	13.1	20.2	26.1	28.9	30.2	28.4	27.4	20.8	12.9	3.4	(23)	
		MCWB	1.2	1.8	10.4	12.0	17.0	20.4	23.2	21.9	19.8	14.3	9.2	2.1	(24)	
	10%	DB	0.4	2.0	9.9	17.4	23.0	27.2	28.7	27.2	25.1	17.5	10.1	2.2	(25)	
		MCWB	-0.2	0.7	7.5	10.7	15.5	19.2	22.2	20.8	18.9	12.5	7.2	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	9.9	16.4	16.8	22.2	25.5	27.3	25.8	24.8	20.5	15.0	8.8	(27)	
		MCDB	6.7	11.9	19.2	24.1	29.1	31.0	31.8	30.5	29.0	23.9	18.4	9.7	(28)	
	2%	WB	2.1	4.0	13.1	15.0	20.2	23.4	25.6	24.2	22.8	17.2	12.2	3.9	(29)	
		MCDB	3.2	5.8	15.0	20.2	25.7	28.5	30.4	28.7	26.5	22.6	15.3	5.3	(30)	
	5%	WB	0.9	2.0	10.1	13.1	18.5	21.8	24.4	23.1	21.3	14.9	9.6	2.3	(31)	
		MCDB	1.8	3.4	13.3	18.2	23.8	26.7	28.7	27.2	25.4	19.8	12.8	3.4	(32)	
	10%	WB	-0.1	0.5	7.3	11.4	17.0	20.5	23.3	21.9	19.8	12.9	7.3	1.1	(33)	
		MCDB	0.5	1.5	9.7	16.7	21.5	25.0	27.4	25.5	24.3	16.8	10.0	1.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.1	8.8	10.6	11.1	10.4	10.7	10.7	11.6	10.5	8.9	7.6	(35)	
		MCDBR	9.0	9.9	13.1	15.6	15.3	12.9	11.7	11.9	13.7	15.6	13.3	9.0	(36)	
	5% WB	MCWBR	7.8	7.9	9.0	8.6	7.1	5.8	5.9	6.1	6.7	8.5	8.8	7.6	(37)	
		MCWBR	8.5	9.4	12.5	13.4	12.4	11.0	10.7	10.7	11.7	12.9	12.8	8.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.261	0.282	0.310	0.362	0.393	0.396	0.403	0.405	0.373	0.319	0.286	0.263	(40)	
		taud	2.449	2.380	2.425	2.336	2.308	2.345	2.333	2.312	2.370	2.521	2.521	2.458	(41)	
	Ebn at Noon		875	911	928	899	877	873	862	846	846	859	837	835	(42)	
		Edh at Noon	69	90	99	119	126	122	122	121	105	78	65	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.77	2.83	3.74	4.61	5.36	5.97	6.40	5.32	4.16	2.68	1.72	1.27	(44)	
	RadStd		0.21	0.31	0.36	0.46	0.43	0.47	0.24	0.40	0.30	0.36	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page